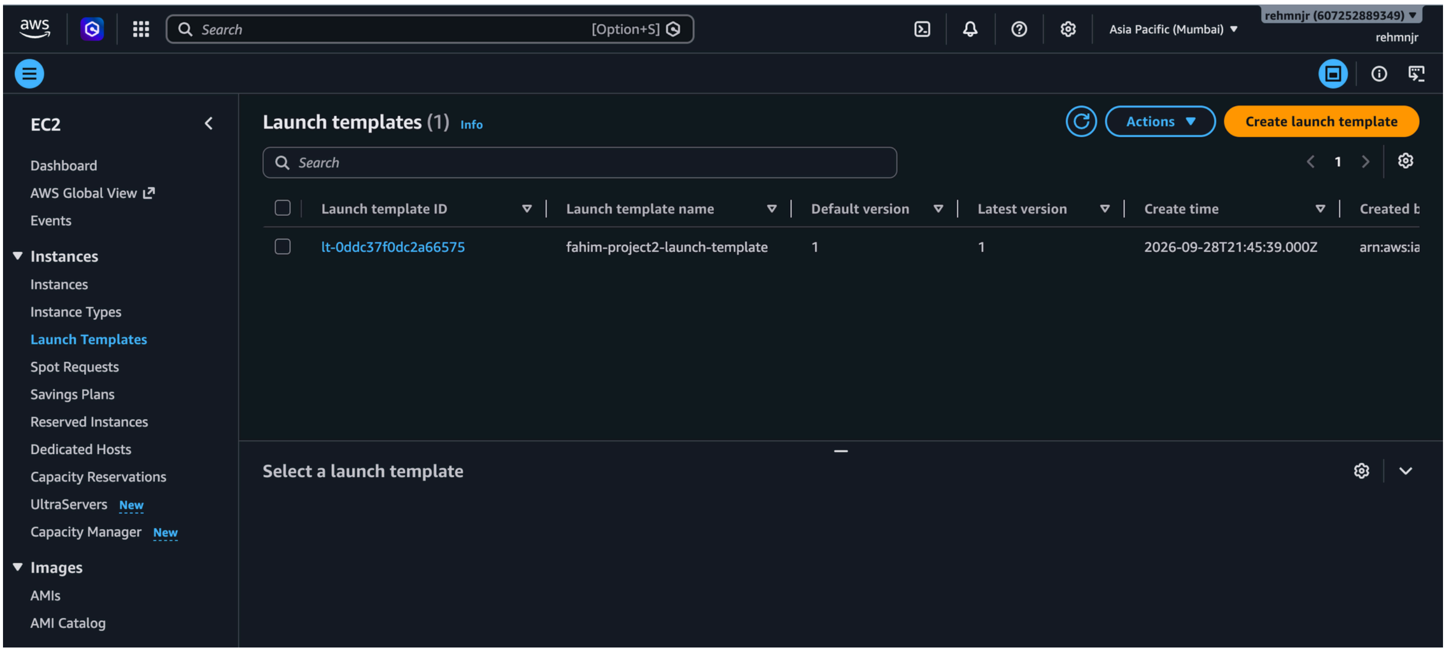
Project 2 – Highly Available Web Application

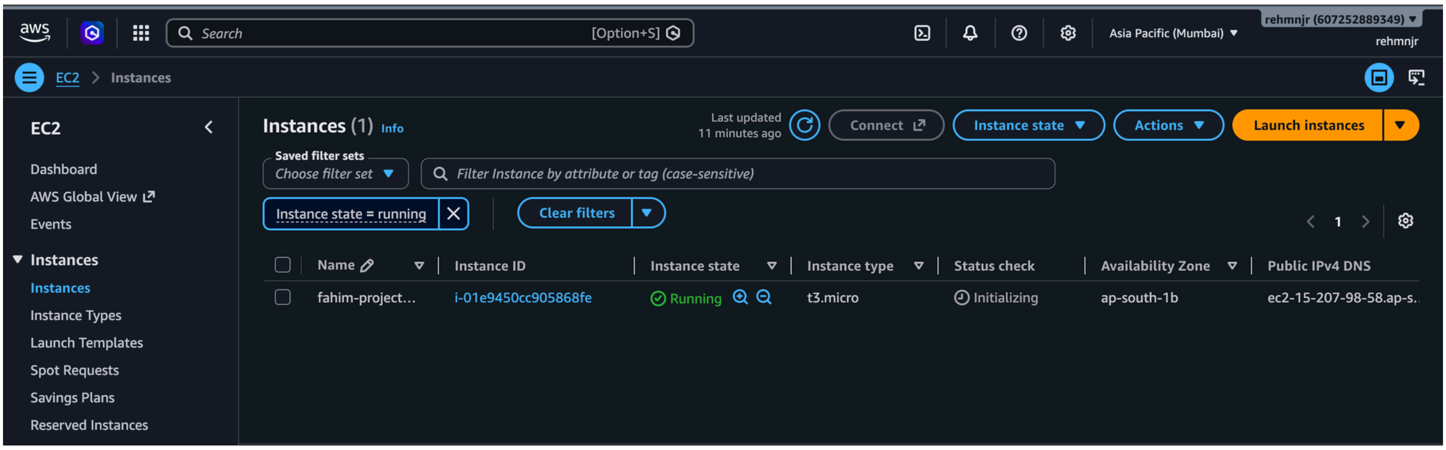
Deploy a highly available web application using EC2, Application Load Balancer, Auto Scaling, EBS, and CloudWatch.

Create a Launch Template



Launch templates (1) Info

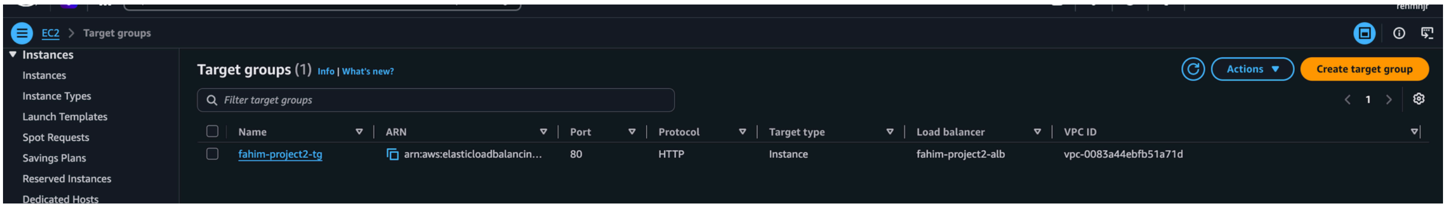
☐	Launch template ID	Launch template name	Default version	Latest version	Create time	Created b
☐	lt-0ddc37f0dc2a66575	fahim-project2-launch-template	1	1	2026-09-28T21:45:39.000Z	arn:aws:ia



Instances (1) Info

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	fahim-project...	i-01e9450cc905868fe	Running	t3.micro	Initializing	ap-south-1b	ec2-15-207-98-58.ap-s..

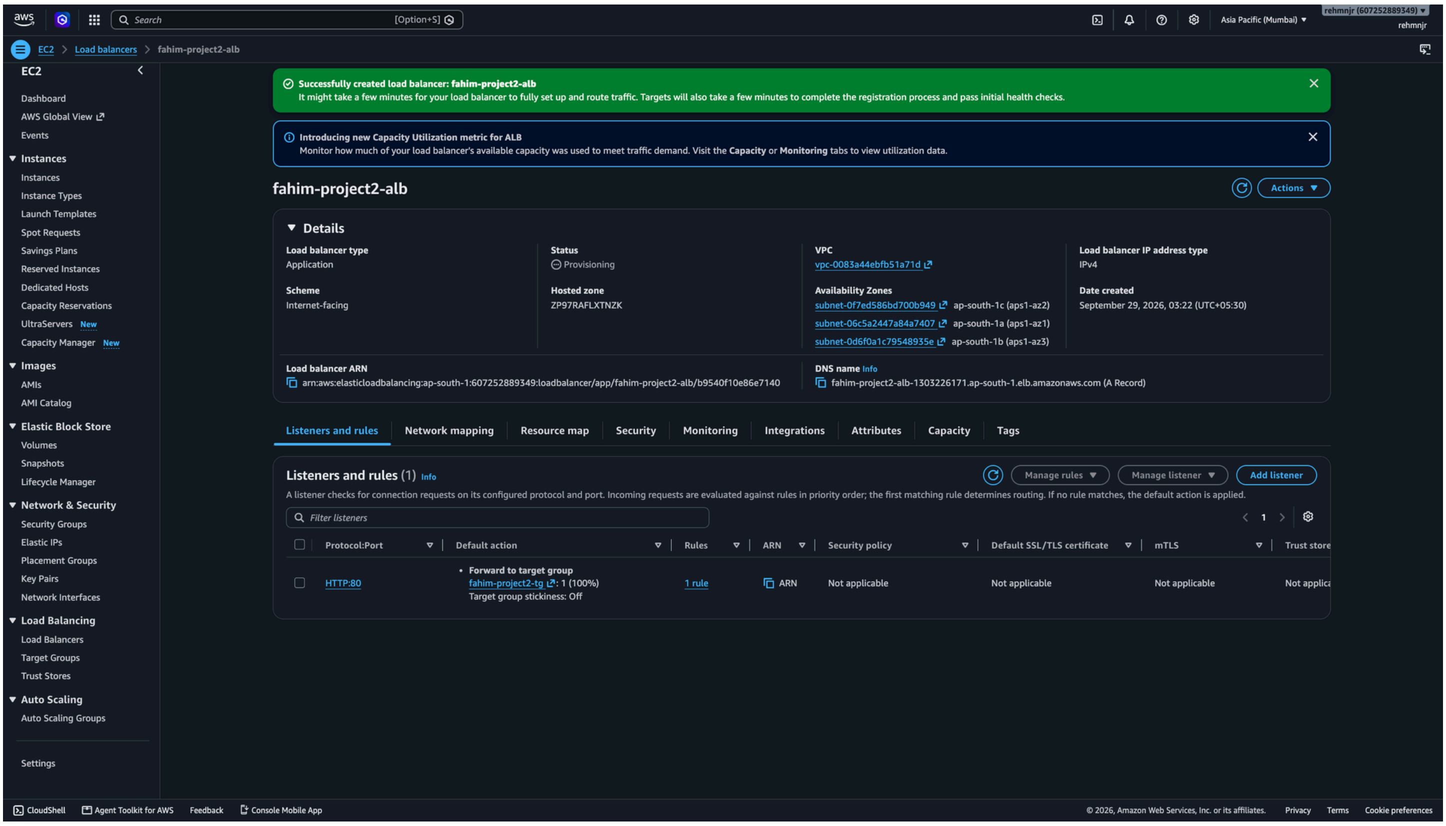
Created Target Group



Target groups (1) Info | What's new?

☐	Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
☐	fahim-project2-tg	arn:aws:elasticloadbalancin...	80	HTTP	Instance	fahim-project2-alb	vpc-0083a44ebfb51a71d

ALB created for HTTP LOAD BALANCING



Successfully created load balancer: fahim-project2-alb
It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

Introducing new Capacity Utilization metric for ALB
Monitor how much of your load balancer's available capacity was used to meet traffic demand. Visit the **Capacity** or **Monitoring** tabs to view utilization data.

fahim-project2-alb

Details

Load balancer type
Application

Scheme
Internet-facing

Status
Provisioning

Hosted zone
ZP97RAFLXTNZK

VPC
vpc-0083a44ebfb51a71d

Availability Zones
subnet-0f7ed586bd700b949 ap-south-1c (aps1-az2)
subnet-06c5a2447a84a7407 ap-south-1a (aps1-az1)
subnet-0d6f0a1c79548935e ap-south-1b (aps1-az3)

Load balancer IP address type
IPv4

Date created
September 29, 2026, 03:22 (UTC+05:30)

Load balancer ARN
arn:aws:elasticloadbalancing:ap-south-1:607252889349:loadbalancer/app/fahim-project2-alb/b9540f10e86e7140

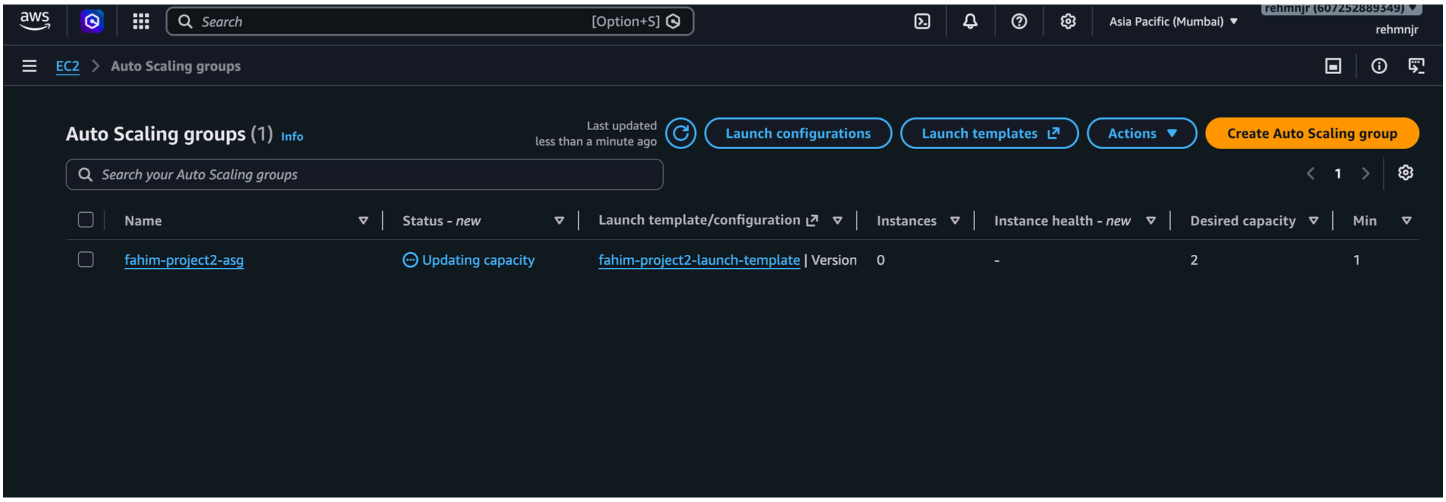
DNS name Info
fahim-project2-alb-1303226171.ap-south-1.elb.amazonaws.com (A Record)

Listeners and rules (1) Info

A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.

☐	Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS	Trust store
☐	HTTP:80	• **Forward to target group** fahim-project2-tg: 1 (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable	Not applica

Created Auto Scaling Group (ASG)

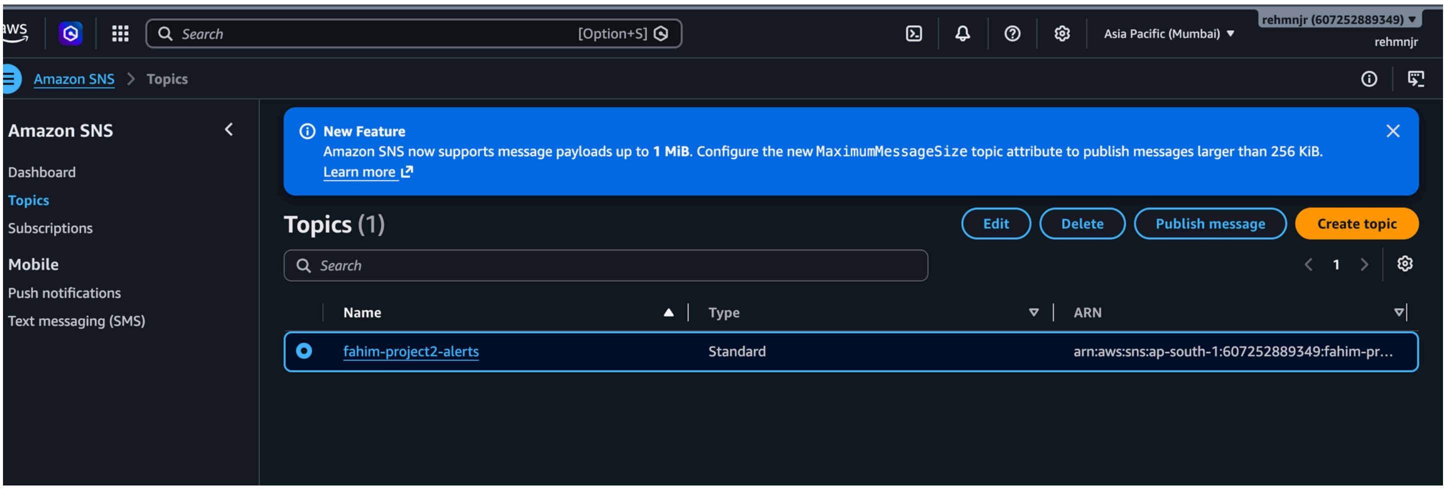


Auto Scaling groups (1) Info

Last updated
less than a minute ago

☐	Name	Status - *new*	Launch template/configuration	Instances	Instance health - *new*	Desired capacity	Min
☐	fahim-project2-asg	Updating capacity	fahim-project2-launch-template | Version	0	-	2	1

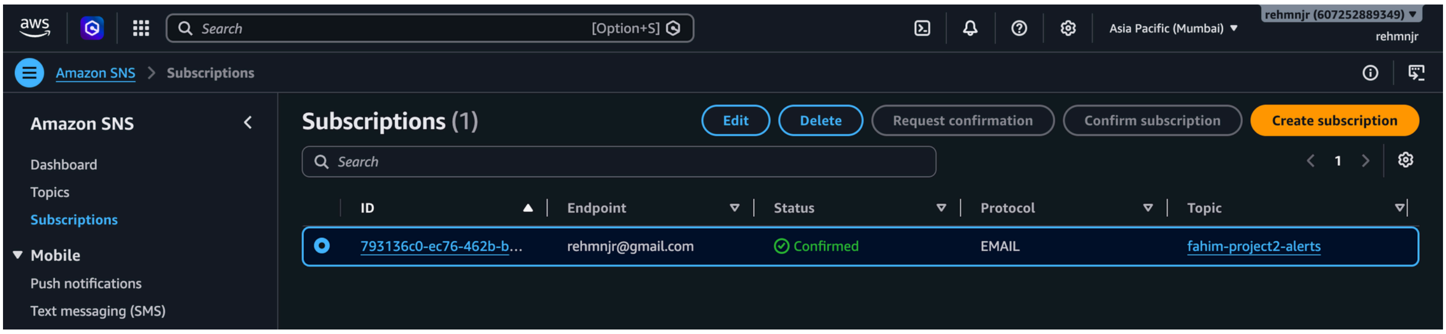
SNS TOPIC Created while creating the ASG



New Feature
Amazon SNS now supports message payloads up to **1 MiB**. Configure the new `MaximumMessageSize` topic attribute to publish messages larger than 256 KiB.
Learn more

Topics (1)

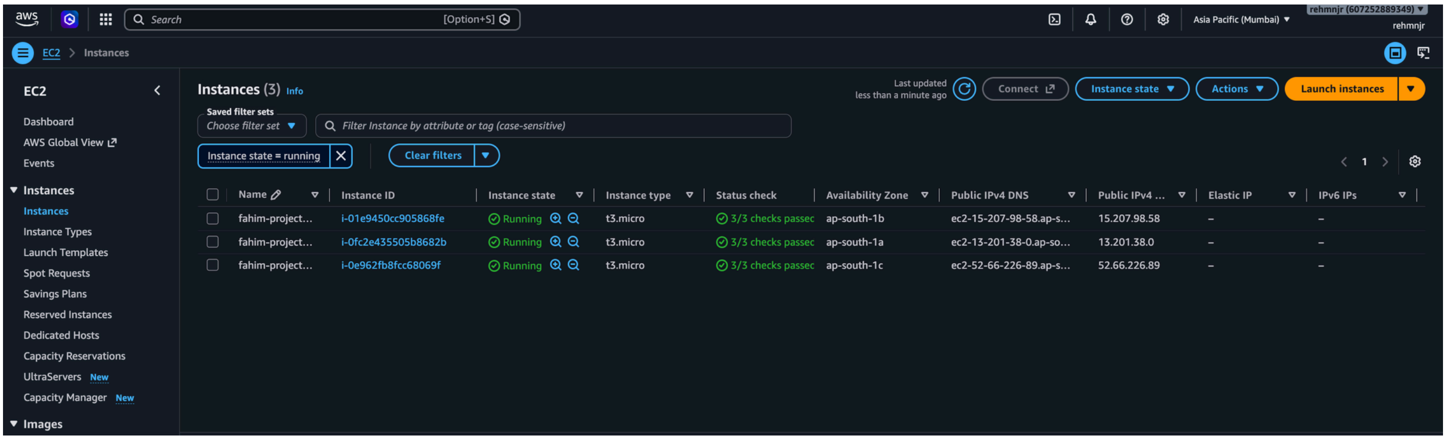
	Name	Type	ARN
●	fahim-project2-alerts	Standard	arn:aws:sns:ap-south-1:607252889349:fahim-pr...



Subscriptions (1)

ID	Endpoint	Status	Protocol	Topic
793136c0-ec76-462b-b...	rehmnjr@gmail.com	Confirmed	EMAIL	fahim-project2-alerts

2 instances automatically created by ASG



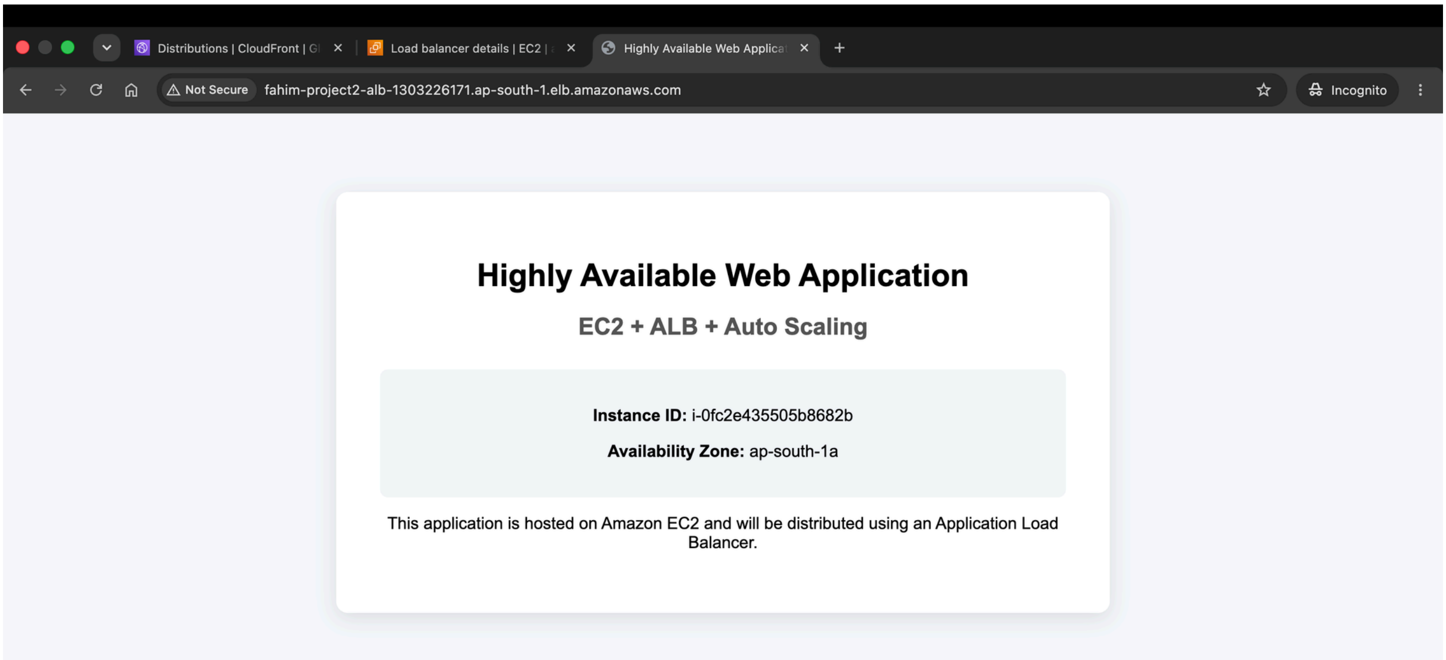
Instances (3)

Instance state = running

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
fahim-project...	i-01e9450cc905868fe	Running	t3.micro	3/3 checks passec	ap-south-1b	ec2-15-207-98-58.ap-s...	15.207.98.58	–	–
fahim-project...	i-0fc2e435505b8682b	Running	t3.micro	3/3 checks passec	ap-south-1a	ec2-13-201-38-0.ap-so...	13.201.38.0	–	–
fahim-project...	i-0e962fb8fcc68069f	Running	t3.micro	3/3 checks passec	ap-south-1c	ec2-52-66-226-89.ap-s...	52.66.226.89	–	–

Testing ALB DNS : http://fahim-project2-alb-1303226171.ap-south-1.elb.amazonaws.com

Instacne ID : `i-0fc2e435505b8682b`


Distributions | CloudFront | G
Load balancer details | EC2 |
Highly Available Web Applica
Not Secure
fahim-project2-alb-1303226171.ap-south-1.elb.amazonaws.com
Incognito

Highly Available Web Application

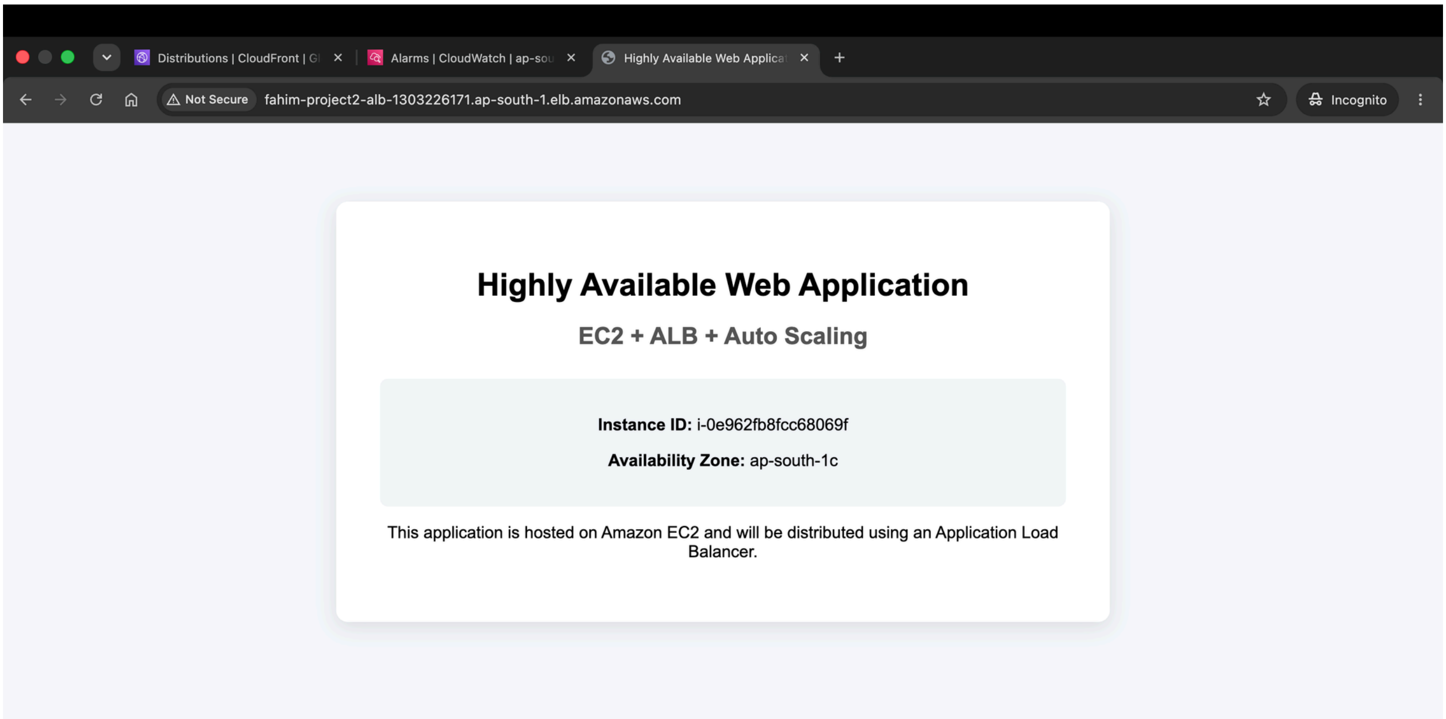
EC2 + ALB + Auto Scaling

Instance ID: i-0fc2e435505b8682b

Availability Zone: ap-south-1a

This application is hosted on Amazon EC2 and will be distributed using an Application Load Balancer.

After Reload Instance ID : i-0e962fb8fcc68069f


Distributions | CloudFront | G
Alarms | CloudWatch | ap-sou
Highly Available Web Applica
Not Secure
fahim-project2-alb-1303226171.ap-south-1.elb.amazonaws.com
Incognito

Highly Available Web Application

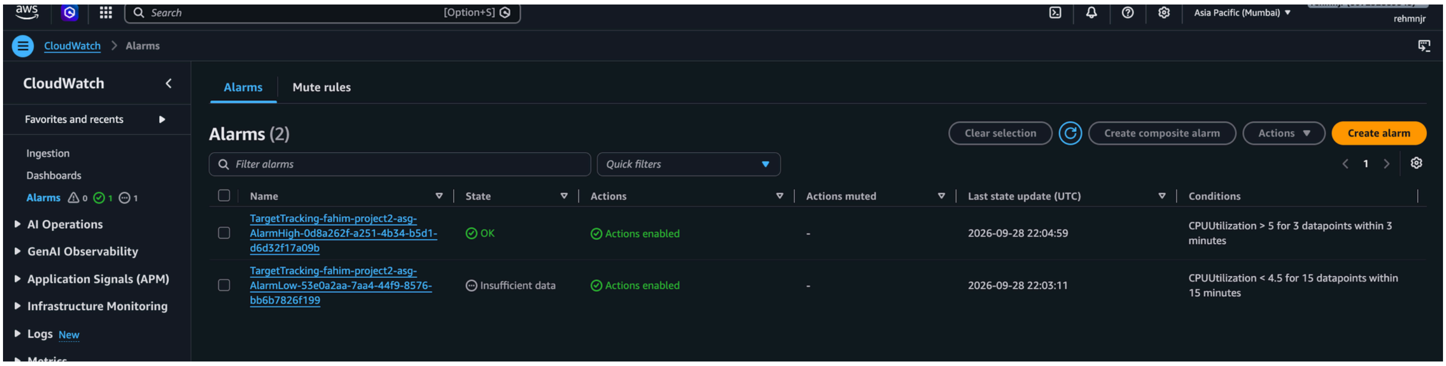
EC2 + ALB + Auto Scaling

Instance ID: i-0e962fb8fcc68069f

Availability Zone: ap-south-1c

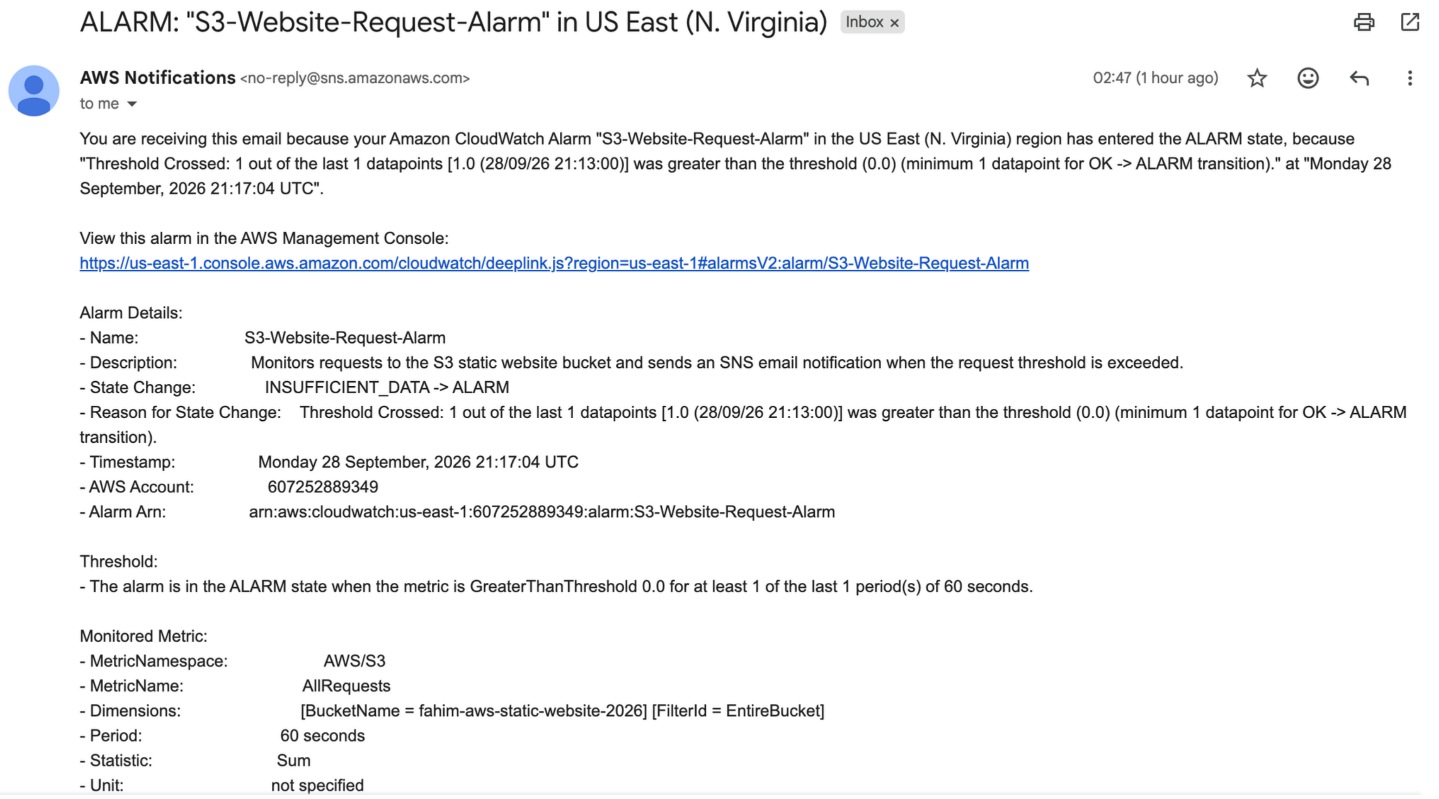
This application is hosted on Amazon EC2 and will be distributed using an Application Load Balancer.

Alarm Generated



Alarms (2)

☐	Name	State	Actions	Actions muted	Last state update (UTC)	Conditions
☐	TargetTracking-fahim-project2-asg-AlarmHigh-0d8a262f-a251-4b34-b5d1-d6d32f17a09b	OK	Actions enabled	-	2026-09-28 22:04:59	CPUUtilization > 5 for 3 datapoints within 3 minutes
☐	TargetTracking-fahim-project2-asg-AlarmLow-53e0a2aa-7aa4-44f9-8576-bb6b7826f199	Insufficient data	Actions enabled	-	2026-09-28 22:03:11	CPUUtilization < 4.5 for 15 datapoints within 15 minutes



ALARM: "S3-Website-Request-Alarm" in US East (N. Virginia)

AWS Notifications <no-reply@sns.amazonaws.com>
to me
02:47 (1 hour ago)

You are receiving this email because your Amazon CloudWatch Alarm "S3-Website-Request-Alarm" in the US East (N. Virginia) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [1.0 (28/09/26 21:13:00)] was greater than the threshold (0.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Monday 28 September, 2026 21:17:04 UTC".

View this alarm in the AWS Management Console:
https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1#alarmsV2:alarm/S3-Website-Request-Alarm

Alarm Details:
- Name: S3-Website-Request-Alarm
- Description: Monitors requests to the S3 static website bucket and sends an SNS email notification when the request threshold is exceeded.
- State Change: INSUFFICIENT_DATA -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [1.0 (28/09/26 21:13:00)] was greater than the threshold (0.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Monday 28 September, 2026 21:17:04 UTC
- AWS Account: 607252889349
- Alarm Arn: arn:aws:cloudwatch:us-east-1:607252889349:alarm:S3-Website-Request-Alarm

Threshold:
- The alarm is in the ALARM state when the metric is GreaterThanThreshold 0.0 for at least 1 of the last 1 period(s) of 60 seconds.

Monitored Metric:
- MetricNamespace: AWS/S3
- MetricName: AllRequests
- Dimensions: [BucketName = fahim-aws-static-website-2026] [FilterId = EntireBucket]
- Period: 60 seconds
- Statistic: Sum
- Unit: not specified

END PROJECT-2